Trustworthy Deep Learning for Chest X-ray Disease Detection: A Model-Agnostic Shortcut-Suppression Module

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Abstract

Deep-learning chest X-ray (CXR) classifiers report high accuracy yet often rely on *shortcut learning*—exploiting hospital-source or acquisition artefacts rather than lung pathology—so accuracy collapses on unseen hospitals, with AUC falling from high values to low values when checking on external X-Rays.

We propose a lightweight, **model-agnostic Shortcut Suppression Module** that can be integrated into any deep learning backbone to encourage attention toward clinically relevant lung regions while suppressing shortcut learning. The proposed module is evaluated through a comprehensive trustworthiness framework encompassing classification performance, calibration, explanation faithfulness, computational efficiency, and cross-hospital robustness across both CNN and Transformer architectures.

1 Problem Motivation

Chest X-ray is the most common medical imaging examination worldwide, and automated screening is especially valuable in settings where radiologists are scarce. However, high headline accuracy can be misleading. Models frequently exploit dataset artefacts, including hospital-source markers, acquisition-device signatures, embedded text, and image borders, rather than genuine lung pathology.

When evaluated on data from previously unseen hospitals, performance can drop substantially due to shortcut learning, with studies reporting large decreases in ROC-AUC despite excellent internal test performance [1]. A CXR model intended for clinical use must therefore satisfy three conditions beyond accuracy:

(1) It must focus on diseased lung regions rather than artefacts.
(2) Its confidence scores must be calibrated so that clinicians can interpret predicted probabilities reliably.
(3) It must generalize to data obtained from previously unseen hospitals.

Why it matters and who benefits. The proposed system can serve as a reliable triage tool in low-resource clinics and mobile screening programmes, helping healthcare providers prioritize patients with suspected chest abnormalities for timely expert review. Radiologists gain a second reader whose reasoning can be inspected, while the wider research community gains a reusable shortcut-suppression technique.

Existing studies address these properties in isolation and offer no lightweight architectural solution that generalizes across model families, which is the gap targeted by this project.

2 Identified Datasets

Primary: COVID-19 Radiography Database (Kaggle) [2]— approximately 21,000 frontal CXRs across four classes: COVID-19, Normal, Viral Pneumonia, and Lung Opacity. The dataset is publicly available, well structured, and Imbalanced classes, making it suitable for studying shortcut learning. Class imbalance is handled using class weighting.

External: RSNA Pneumonia Detection (Kaggle) [3]—frontal CXRs labelled as Pneumonia and Normal from different hospital sources. This dataset is used *exclusively* for out-of-distribution robustness evaluation. Models are never trained using this dataset; instead, the internal-to-external accuracy and AUC drops are used as robustness measures.

3 Baseline Model

3.1 Model Architecture

A Convolutional Neural Network (CNN) is adopted as the baseline architecture, as CNNs have been widely used for image classification and provide a strong benchmark against which more advanced models can be evaluated.

DenseNet121 [4] is selected as the primary model due to its established medical image classification performance, dense feature reuse, and relatively compact architecture of approximately 8.0 million parameters. The model is initialized using ImageNet-pretrained weights, and its final classifier is replaced with a four-class classification head.

3.2 Implementation Details

Images are resized to 224×224, converted to three channels, and normalized using ImageNet statistics. The dataset is split into training, validation, and test sets (70/15/15) using stratified sampling with a fixed random seed.

To improve generalization, training images are augmented using random rotations, horizontal flips, brightness/contrast adjustments, and random resized cropping. The model is implemented in PyTorch using the `timm` library, optimized with `AdamW`, and trained using class-weighted cross-entropy loss.

A two-phase transfer-learning strategy is adopted: first, the backbone is frozen while training the classification head; then, the final dense blocks are unfrozen for fine-tuning with a reduced learning rate. Early stopping based on validation loss is applied to reduce overfitting. Experiments are conducted using Google Colab and Kaggle GPU environments.

3.3 Baseline Experimental Results and Hyperparameter Tuning

Default hyperparameters may provide limited performance. Therefore, systematic hyperparameter tuning is performed over the search space shown in Table 1. Table 2 reports the resulting baseline performance for the frozen and fine-tuned models trained on original data without applying lung masking. The values will be populated using the Week 1 experimental results and finalized before submission.

Table 1: Hyperparameter search space for DenseNet121.

Hyperparameter	Values searched
Initial learning rate	$\{1 \times 10^{-3}, 3 \times 10^{-4}, 1 \times 10^{-4}, 3 \times 10^{-5}\}$
Batch size	$\{16, 32, 64\}$
Optimizer	Adam, AdamW, SGD with momentum
Weight decay	$\{0, 1 \times 10^{-4}, 1 \times 10^{-3}\}$
Learning-rate scheduler	Cosine, step decay, reduce-on-plateau

Table 2: DenseNet121 baseline experimental results after hyperparameter tuning.

Setting	Accuracy	Macro-F1	ROC-AUC
Phase 1 (Frozen Head)	85.80%	85.61%	97.19%
Phase 2 (Fine-tuned)	**91.59%**	**91.72%**	**98.71%**

4 Proposed Contribution

4.1 Expected Novelty

We propose a lightweight, **model-agnostic Shortcut-Suppression Module** that can be inserted into different backbone architectures to steer model attention toward lung regions. Three candidate designs are initially explored:

(1) **Lung-Region Attention Module:** A spatial-attention block guided by an automatically generated lung mask. The module reweights feature maps to suppress background regions and imaging artefacts.
(2) **Auxiliary Segmentation Head:** A secondary decoder that jointly predicts a lung mask during training, thereby acting as an anatomical regularizer.
(3) **Shortcut-Suppression Loss:** A regularization term that penalizes Grad-CAM activations located outside the lung regions.

The best-performing design is then selected and evaluated across four architectures: DenseNet121, ResNet50, EfficientNet-B0, and ViT-Base [4–7].

This evaluation is intended to demonstrate whether the proposed solution generalizes across both CNN and Transformer model families. The proposed lightweight shortcut-suppression framework enables consistent trustworthiness evaluation across both architectures.

4.2 Expected Improvement

The proposed module is expected to achieve the following improvements:

(1) Lung Masking is implemented in-order to extract and isolate the lung regions in chest x-rays.
(2) A reduced internal-to-external accuracy and AUC drop (internal AUC Score - external AUC Score), indicating improved robustness.
(3) A higher lung-localization faithfulness score, indicating greater attention to lung regions rather than artefacts.
(4) Maintained in-distribution classification performance without a significant reduction in accuracy.
(5) Negligible parameter and computational overhead, allowing the model to remain deployable in resource-constrained environments.

The controlled comparison between models trained with and without the proposed module provides direct evidence of whether the module reduces shortcut reliance or merely shifts the learned shortcuts to different image regions.

5 Experimental Plan

5.1 Evaluation Metrics

The models are evaluated across five dimensions:

- **Classification performance:** Accuracy, precision, recall, F1-score, ROC-AUC, per-class metrics, macro-averaged metrics, and confusion matrices.
- **Calibration:** Expected Calibration Error, Brier score, and reliability diagrams before and after temperature scaling [8].
- **Explanation faithfulness: Grad-CAM** [9] for CNN architectures and attention rollout for the Transformer architecture, together with a quantitative lung-localization score.
- **Efficiency:** Parameter count, FLOPs, model size, and CPU/GPU inference latency.
- **Robustness:** The difference between internal and external accuracy and ROC-AUC.

5.2 Baselines

The primary baseline is the standard DenseNet121 architecture. ResNet50, EfficientNet-B0, and ViT-Base are used as additional baselines.

All architectures are trained using an identical experimental pipeline, including the same preprocessing, augmentation, optimizer, dataset split, and random seed. This ensures that observed performance differences can be attributed to the model architecture and the proposed shortcut-suppression module.

Frozen and fine-tuned transfer-learning settings are compared for each architecture. Every model is evaluated both with and without the proposed module.

5.3 Resource Requirements

All experiments are conducted using freely available Google Colab and Kaggle GPU environments. Images are processed at a resolution of 224×224, and mixed-precision training is used to reduce memory and computational requirements.

No multi-GPU clusters or high-performance computing infrastructure are required. Lung masks are generated using an off-the-shelf lung segmentation model to ensure that the experimental pipeline remains feasible within the one-month project duration and available GPU limits.

References

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